

ME 333 - MECHANICS OF MATERIALS
PRACTICE PROBLEMS 2 - CONCEPT OF STRESS - SEP 11, 2025

Practice Problems:

From Prescribed Text - 2.1, 2.5, 2.6, 2.7, 2.11, 2.15, 2.17, 2.27.

- (1) Given are the direction cosines (l, m, n) with respect to (x, y, z) of the normal to an arbitrary plane P on which the traction vector is $\boldsymbol{\sigma}_P$ and the normal stress is σ_{PN} . Determine the direction cosines of the in-plane resultant shear σ_{PS} also with respect to (x, y, z) .

[Answer] $l_1 = \frac{\sigma_{Px} - l\sigma_{PN}}{\sigma_{PS}}; \quad m_1 = \frac{\sigma_{Py} - m\sigma_{PN}}{\sigma_{PS}}; \quad n_1 = \frac{\sigma_{Pz} - n\sigma_{PN}}{\sigma_{PS}}$

- (2) The components of stress tensor at a point in a solid (in MPa) are $\sigma_{xx} = 6, \sigma_{yy} = 6, \sigma_{zz} = 8, \tau_{xy} = -3$ and $\tau_{xz} = \tau_{yz} = 0$. Then determine:

- (a) the principal stresses and the direction cosines of the corresponding planes
- (b) the normal and shear components of the stress vector on a plane with normal $(2/3, -2/3, 1/3)$
- (c) the invariants
- (d) the principal deviatoric stresses
- (e) the invariants of the deviatoric stress tensor

[Answer] (a) $(9, 8, 3)$ and $\{-1/\sqrt{2}, 1/\sqrt{2}, 0\}, \{0, 0, 1\}, \{-1/\sqrt{2}, -1/\sqrt{2}, 0\}$; (b) $(8.9, 0.3414)$ MPa; (c) $(20, 123, 216)$; (d) $(-3.67, 1.33, 2.33)$; (e) $(0, -10.33, -11.41)$.

- (3) The stress state at a point in a solid (in MPa) is given by: $\sigma_{xx} = p, \sigma_{yy} = p, \sigma_{zz} = p, \tau_{xy} = ap$ and $\tau_{xz} = bp, \tau_{yz} = cp$, where (a, b, c) are constants and p is some stress value. Determine the constants so that the stress vector vanishes on the Octahedral plane.

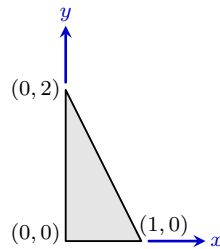
[Answer] $p = 0 \forall (a, b, c)$

- (4) At a point P , the principal stresses are such that $2\sigma_2 = \sigma_1 + \sigma_3$. Determine the unit normal $\mathbf{N}$ for the plane on which the normal and shear stresses are : $\sigma_N = \sigma_2$ and $\sigma_S = (\sigma_1 - \sigma_3)/4$.

[Answer] $(1/2\sqrt{2}, \sqrt{3}/2, 1/2\sqrt{2})$

- (5) Compute the tractions on the faces of a right triangular body as shown in Figure. 5, considering the given two dimensional stress state

$$\sigma_{xx} = 0; \quad \sigma_{yy} = C_1 y; \quad \tau_{xy} = -C_1 x$$



[Answer] Left face: $\mathbf{n} = \begin{Bmatrix} -1 \\ 0 \end{Bmatrix}, \mathbf{t} = \begin{Bmatrix} 0 \\ C_1 x \end{Bmatrix}$; Bottom face: $\mathbf{n} = \begin{Bmatrix} 0 \\ -1 \end{Bmatrix}, \mathbf{t} = \begin{Bmatrix} C_1 x \\ -C_1 y \end{Bmatrix}$; Inclined face: $\mathbf{n} = \frac{1}{\sqrt{5}} \begin{Bmatrix} 2 \\ 1 \end{Bmatrix}, \mathbf{t} = \begin{Bmatrix} -C_1 x/\sqrt{5} \\ (-2C_1 x + C_1 y)/\sqrt{5} \end{Bmatrix}$

- (6) The stresses acting at a point in a piece of material are $\sigma_{xx} = 4$, $\sigma_{yy} = 7$, $\sigma_{zz} = -9$, $\tau_{xy} = 4$ and $\tau_{xz} = 0$, $\tau_{yz} = 5$. Calculate:
- (a) the normal stress, resultant stress and the shear stress on a plane whose normal makes angles of 50° with the positive x -axis and 60° with the positive y -axis.
 - (b) the direction cosines of the shear stress with respect to the x , y and z -axes.
 - (c) the principal stresses and the direction cosines of the normals to the principal planes with respect to x , y and z .
 - (d) the maximum shear stress at the point in the material.

[Answer] (a) (5.84, 8.64) MPa; (b) (0.09426, 0.70023, -0.70767); (c) (10.67, 1.85, -10.52) MPa and (-0.5025, -0.8379, -0.213), (0.8609, -0.4621, -0.2129), (-0.08, 0.2904, -0.9536); (d) 10.59 MPa